

KRISHNA RANI

AI Engineer · M.Sc. Computer Science

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PROFILE

AI Engineer completing an M.Sc. in Computer Science at Leibniz Universität Hannover in February 2027, with a focus on LLMs, NLP, model evaluation, and AI-driven data pipelines. Current work includes LLM benchmarking and ontology learning, alongside 4+ years of industry experience in software engineering at Bosch.

TECHNICAL SKILLS

Programming	Python, C, C++
ML / NLP	PyTorch, Hugging Face Transformers, Sentence Transformers, LLMs, RAG
Embedded / Audio	Linux, Audio HAL, Audio Middleware
Tools / DevOps	Git, Gerrit, ClearCase, Jenkins, Google Test, PyPI Packaging
HPC	NVIDIA H100/L40S GPU Compute, SLURM Job Scheduling

ACADEMIC EXPERIENCE

12/2024 – Present	Student Assistant SciKnowOrg Group, TIB – Leibniz Information Centre for Science and Technology, Hannover, Germany - Assist with maintaining the codebase of OntoLearner and improving existing modules of OntoAligner, open-source Python libraries for LLM-based ontology learning and alignment, for the smooth operation of both projects. - Integrated the top 4 LLMs4OL Challenge (ISWC) solutions into OntoLearner as 11 reusable learner models across term typing, taxonomy discovery and Text2Onto, generalizing dataset-specific research code to work with any ontology in the library. - Compiled and processed PubMed paper metadata in virology and epidemiology into a dataset for biomedical literature mining, using Python and Jupyter Notebooks.
03/2026 – 09/2026	Master Thesis: LLM Benchmarking in High School Education Leibniz Universität Hannover, Hannover, Germany · doi.org/10.15488/22429 - Built a PDF-to-JSON pipeline using Chandra/DeepSeek-OCR and Qwen3-32B, processing exams from 25 sources through configurable preprocessing, automated checks and manual review. - Delivered HighSchoolBench, a multimodal benchmark of 10,501 Grades 9–12 exam questions across 11 subjects, stored as structured JSON with linked images and tables for automated model evaluation. - Developed an evaluation tool for 22 models, including Qwen, Mistral, DeepSeek and GPT-OSS, on 8,393 test questions using OpenAI-compatible APIs, SLURM GPUs (H100/L40S) and automated answer scoring.

WORK EXPERIENCE

- 09/2019 – 04/2024

Senior Software Engineer

Robert Bosch, Bangalore, India

 - Owned external amplifier software for Mitsubishi Android infotainment systems, covering MCAN, Bose and Yamaha components across development, integration, testing and issue resolution; later owned the volume module.
 - Developed C++ Audio Manager features for audio-source prioritisation, mixing-channel handling, attenuation and volume configuration.
 - Adapted and validated audio software for replacement Texas Instruments amplifier chips in Nissan and Renault projects, enabling hardware compatibility during the semiconductor shortage.
 - Automated Google Test line-coverage reporting with a Python tool, saving the quality management team an estimated 100 hours of manual effort per month.
- 10/2022 – 01/2023

Client Assignment: Senior Software Engineer

Robert Bosch GmbH, Hildesheim, Germany

 - Supported Nissan and Renault Gen3/Gen4 infotainment audio projects through bug analysis and resolution of production-plant and customer-reported issues.

EDUCATION

- 05/2024 – 03/2027

M.Sc. in Computer Science

Leibniz Universität Hannover, Hannover, Germany · Current grade: 1.6
- 06/2015 – 06/2019

Bachelor of Engineering in Computer Science

Bangalore Institute of Technology, Bangalore, India · Final grade: 2.3

ACADEMIC PROJECTS

- FedLIME: Federated Learning with LIME

GitHub

 - Built a federated learning pipeline with Flower (FedAvg) that trains a shared neural network across 3–10 simulated clients, using both random and demographic-based data splits.
 - Generated LIME explanations locally on each client and shared only noise-protected summaries (Gaussian local differential privacy), so no raw data or detailed explanations leave the clients.
 - Evaluated on 3 datasets (Adult Income, Bank Marketing, Law School): federated explanations reached up to 0.96 similarity to the centralized baseline and matched or exceeded its fidelity (up to 0.53 vs. 0.30 R^2).
- RNA-to-Protein Prediction Using Machine Learning and Deep Learning Models

GitHub

 - Predicted protein expression from single-cell RNA data using the Kaggle Open Problems – Multimodal Single-Cell Integration dataset.
 - Trained and compared Ridge Regression, Random Forest, a fully connected neural network and an autoencoder as baseline models.
 - Built a prediction pipeline on the pretrained scGPT single-cell foundation model, adding a neural network head to predict protein levels.

LANGUAGES

- English

Fluent
- German

B1